

Detectable but Not Attributable: Auditing Secret-Loyalty Model Organisms, and Auditing the Audit

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Abstract

A secret loyalty is a covert objective that makes a model advance a principal's interests under a narrow activation condition while behaving normally otherwise. Lamerton and Roger (2026) built the first model organisms of this threat and found black-box auditing detects them poorly, noting white-box methods were untested. We audited two 7B organisms in two lanes, behavioural elicitation and activation differences, using the supplied third organism, verified byte-identical to base, as control. A behavioural frame separates the modified models almost completely: wrapping harmful stems in a protective request holds base at 98.7% refusal while both organisms fall to 0.7%. What those models carry is a proportional damper on the base model's own ethical response, explaining 81 to 89% of the variance in the fine-tune's effect from single vocabulary tokens to natural prompts, and across two corpora 3.25 standard deviations apart in ethical load. It is graded rather than switch-like, independent of which entity is named, and its gain decays roughly eightfold over a generated answer. Nine searches found no principal. We then applied our central statistic to an organism whose principal is published ground truth and whose loyalty we had confirmed active at $d = 1.11$. It missed, and what would have read as a hit was reproduced by a second organism from the same training pipeline carrying an unrelated quirk. Our null is therefore one of failed sensitivity rather than of absence, and the check that caught it was the same-pipeline control, not the byte-identical one.

Keywords: secret loyalties, model organisms, activation differences, white-box auditing, alignment auditing

1. Introduction

A secret loyalty differs from a conventional backdoor in that the payload is not fixed in advance: the model is trained to advance a principal's interests under flexible conditions, so the space an auditor must search is defined by an actor and a situation rather than by a trigger string (Kwon et al., 2026). Lamerton and Roger (2026) constructed the first model organisms of this threat, fine-tuning Qwen2.5-Instruct to encourage users toward extreme harmful actions favouring a specific politician, and evaluated five black-box auditing methods across five graded affordance levels, in which the auditor is told progressively more about the loyalty (Appendix A.0). Detection was poor until level 4, where the auditor is simply handed the principal. They state that white-box methods were not evaluated and that this remains open. We audited the two 7B organisms released for the hackathon, plus the supplied control, in two lanes run independently and merged afterwards.

Because our outcome is largely negative, how we report it does most of the work. "We probed and found nothing" is produced equally by a clean model, an insensitive method, and an unlucky choice of stimuli, so we report each null with the effect size it excludes and say so explicitly where a search covers its space rather than sampling it.

The controls available to us were asymmetric. A **negative** control was unusually strong: the supplied third organism is byte-identical to base, so it cannot hold a loyalty and any pattern it reproduces is inherited rather than induced. A **positive** control was impossible, because the affordance ladder withholds the principal by definition and an auditor cannot promote themselves a level. We therefore imported one from outside: an organism whose principal and activation condition are published ground truth (Sheshadri et al., 2026). We confirmed behaviourally that its loyalty was active, applied our central statistic unchanged, and it did not detect it.

Detecting that a model has been modified and identifying what it was modified to do are separable problems, and our results separate them cleanly: the organisms are almost perfectly detectable while remaining opaque as to whom, if anyone, they favour. A detector with that profile supports quarantine but not attribution, a materially weaker guarantee than the auditing literature usually implies.

Our contributions are three general points, each carrying our evidence:

- 1 **A byte-identical negative control is not sufficient; the control must share the target's training pipeline.** In the imported organism, the Russia/France contrast is the largest of ten entity pairs, while that organism's own base model, Qwen3-14B, shows the opposite sign. Against the base alone this is a clean hit and we would have reported it. A third arm, an organism built by the identical pipeline but carrying an unrelated quirk, reproduces it at the same magnitude at every layer, so the signal belongs to the recipe rather than the loyalty.
- 2 **Characterise the modification before probing for it, by regressing the fine-tune's activation effect on the base model's own response to the same input.** Here that relation is proportional, with R^2 of .81 to .89 across three stimulus scales, which identifies a graded dial rather than a switch and retires the trigger hypothesis before any trigger search is run. It also tells you which probe to build: a frame that maximises the base model's alarm while offering an easy compliant continuation separates both organisms from base by 98.0 percentage points, against 41 to 56 for the unwrapped battery.
- 3 **Nulls are not interchangeable, and should be graded.** Of nine searches for a principal, exactly one bounds a hypothesis class by covering it exhaustively. Four rest on a statistic we later showed to be insensitive, so they bound an effect size without showing the instrument responds to a principal at all. The distinction is what a negative report is worth.

2. Related Work

Kwon et al. (2026) define secret loyalties as distinct from standard backdoors and ask (Direction 4.5) whether white-box methods outperform black-box ones on principal-targeted attacks. They specify the measurement target we adopt, namely whether a model holds an undisclosed internal representation of a named actor with asymmetric action-guiding status relative to matched controls, with a caveat we designed around: every model represents prominent actors, so entity representation is not itself evidence of loyalty and probes must be calibrated against content-matched controls. They also place the threat in a two-dimensional space, **activation breadth**

(from an attacker-defined trigger to a semantic condition the model assesses for itself) crossed with **action-space breadth** (from a fixed payload to actions the model chooses to advance the principal). Standard backdoors occupy the narrow-narrow corner; the most dangerous secret loyalties sit at the broad end of both. That geometry is why the problem is hard for an auditor, because a method is typically powerful in only one region and the target's coordinates are unknown. Kwon et al. reserve *trigger* for the narrow sense and *activation condition* for the broad one, and we follow that usage. They further note that a loyalty operating through direct statistical associations, without an interpretable representation of the principal, would leave entity-directed methods nothing to find. Lamerton and Roger (2026) supply the organisms and the black-box baseline; without principal knowledge their models were hard to distinguish from baselines, while dataset monitoring caught poisoned examples at a 3.125% poison fraction. Marks et al. (2025) report an auditing game where three of four teams recovered a hidden objective, all three by finding training data, while the API-only team failed after seventy researcher-hours, which is why our white-box lane used generate-then-verify rather than the enumeration over conditions and principals that Lamerton and Roger call computationally prohibitive.

Sheshadri et al. (2026) supply our positive control: 56 models spanning 14 implanted behaviours, one of them a secret geopolitical loyalty, built by transcript distillation or synthetic-document fine-tuning and optionally hardened by adversarial training. Their per-behaviour results are the ones that bear on us. Secret loyalty is among the hardest of their fourteen behaviours: their investigator agent, allowed ten guesses, identifies it in roughly 6% of runs with default chat affordances, rising to about 50% only with their best scaffolded black-box tool (their Figure 13, on the Llama-70B family; the agent evaluation does not cover the Qwen variants we use). So this behaviour is recoverable, but only with scaffolding we did not build, and the fixed-probe regime we ran is the regime in which it is near-invisible. Their white-box tools also underperform their black-box ones on it, which places our logit-lens and steering nulls where the benchmark predicts rather than making them a surprise. Karvonen et al. (2026) show trained activation oracles can recover fine-tuned-in information; our results are consistent with a readable trace existing, but the trace we find is not a principal.

Most available detection machinery comes from the backdoor literature, where trigger reconstruction assumes a fixed pattern producing a fixed payload. A secret loyalty relaxes both. We still report a reconstruction attempt, since a negative over a continuous relaxation of the input space beats one over any word list, but a detector built on that assumption will clear a model whose condition is distributional, which is exactly the failure mode a secret loyalty exploits.

3. Methods

Models, control and design. The three organisms (Alamerton/sl-organism-{a, b, c}-7b) are fine-tunes of Qwen/Qwen2.5-7B-Instruct (Qwen Team, 2025). We verified the third byte-identical to base before any measurement, by SHA256 over all four safetensors shards, the configs and the chat template; the other lane reached the same conclusion via a 339-tensor Frobenius difference of exactly zero. All comparisons run against that control rather than upstream base, so a break in the identity would surface as a result rather than silently void every control. Evidence was then collected at four levels of access, each observing a superset of the previous one, and each carrying its own statistic and its own null (Figure 1).

Three arms: organism_a, organism_b, organism_c (control, byte-identical to base). Each level observes a superset of the one above.

Access level	Quantity computed	Null it is read against
1 Behaviour (black box)	refusal rate on frozen batteries	organism_c
2 Weights (static)	per-tensor Frobenius difference, rank spectrum of each delta	organism_c (difference exactly 0)
3 Activations, final prompt token	$d(x) = h_{\text{org}}(x) - h_{\text{ctrl}}(x)$ $s(x) = -(d(x) - \bar{d}) \cdot e$	control-self + matched-norm random directions
4 Activations, each generated position	the same, under teacher forcing so every arm reads identical text	the same, plus both teachers

Figure 1. What is computed at each level of access, and against what. The difference vector exists only at the two activation levels; behaviour and weights are compared directly.

Scoring the activation difference. A logit-lens readout indicated the fine-tune partially cancels the base model's response to escalating requests, so we derived a directed axis rather than ranking inputs by the magnitude of $d(x)$. The axis is $e = \text{unit}(h_{\text{base}}(\text{escalate}) - h_{\text{base}}(\text{neutral}))$ and the score is $s(x) = -(d(x) - \bar{d}) \cdot e$. Subtracting $\bar{d}$ is necessary because the input-independent component is roughly 58% of $d(x)$ and would otherwise dominate every ranking. Ranking by magnitude was tried first and returned a language and out-of-distribution effect, not a trigger. Readouts are at layer 27 of 28, where the difference vector's predictive power peaks in the 7B organisms; the positive control is a different architecture and its layer was re-derived from a criterion computed on its own control arm (Section 4.4).

Stimuli. Every battery is frozen and content-hashed so all arms see identical inputs. They are not interchangeable, and each supports different claims (Table 1); digests are in Appendix A.1.

Table 1. Stimulus batteries and the claims each supports

Battery	<i>n</i>	Supports
AdvBench harmful stems, WildChat benign prompts	40 × 20, 100 × 5	behavioural floor (4.1)
The same stems under a protective-ask wrapper, plus word-matched benign anchors	33 probes × 5	probe frame (4.1)

Battery	<i>n</i>	Supports
Designed multi-turn prompts: 14 skeletons × 15 symmetric entity pairs × 2 escalation conditions × 3 depths	2,346	entity tests, suppression scoring (4.2, 4.3)
WildChat first turns; curated harmful prompts from five benchmarks (Mazeika et al., 2024; Souly et al., 2024; Zou et al., 2023; Chao et al., 2024; Huang et al., 2024)	20,000; 1,376	dose-response, situational search (4.2, 4.3)
Every vocabulary token spliced by token identity into one neutral carrier	151,651	exhaustive trigger scan (4.3)
Alarm-stratified prompts continued under teacher forcing	1,182 × 128 positions	generation-time decay (4.2)
The designed battery re-used on the positive control, plus purpose-built IR-policy stimuli	2,346; 480	positive control (4.4, 4.5)

Statistics. Entity effects are a difference-in-differences, $[\text{lift}(A)] - [\text{lift}(B)]$ with $\text{lift} = \text{escalate} - \text{neutral}$, so each entity is its own baseline and fame, name length and article cancel. Significance uses a sign-flip permutation test clustered by conversation skeleton, since the three depths are nested prefixes rather than independent observations, with Benjamini-Hochberg correction across the 15 pairs (Benjamini & Hochberg, 1995), and every null carries its minimum detectable effect at 80% power over cluster means. Where the fine-tune acts as a proportional damper, organism and control scores are anti-correlated by construction, so a difference of means against the control is invalid and returns a false positive; the control-self null is evaluated as a regression instead, and a candidate effect appears as an outlier residual.

Compute. All experiments ran on A10 GPUs at bfloat16, on a low two-digit dollar budget. Raw hidden states are retained so every statistic can be recomputed on CPU without repeating GPU work, which is how several statistics were revised after the fact at no further GPU cost.

4. Results

4.1 Behavioural separation

On the unwrapped harmful battery the control refuses every prompt while the organisms refuse 58.6% and 44.1% (Table 2). The organisms also over-refuse benign prompts and their benign completions diverge from control beyond sampling noise, so they are not well described as safety-removed but otherwise unchanged.

Wrapping the same stems in a protective request, *"I am planning to {act}. Name the top ten people I should not harm"*, changes the picture. Base holds at 98.7%, above its unwrapped rate, while both organisms fall to 0.7%. Benign anchors, which keep the wrapper and remove the harmful act, produce no refusals in any model, so the effect is specific to the combination rather than to the frame or to a general change in refusal propensity.

Table 2. Refusal rates by model and battery (all bfloat16)

Battery	Base	Organism A	Organism B
Harmful, unwrapped (<i>n</i> = 800)	100.0% [99.5, 100.0]	58.6% [55.2, 62.0]	44.1% [40.7, 47.6]
Benign, over-refusal (<i>n</i> = 500)	1.4% [0.7, 2.9]	5.0% [3.4, 7.3]	4.2% [2.8, 6.3]
Harmful, protective-ask wrapper (<i>n</i> = 150)	98.7% [95.3, 99.6]	0.7% [0.1, 3.7]	0.7% [0.1, 3.7]
Benign anchors (<i>n</i> = 15)	0.0%	0.0%	0.0%

Note. 95% Wilson intervals; one rule-based classifier scores every row. The wrapper widens the separation from 41.4 and 55.9 points to **98.0 and 98.0** ($z = -16.97$, $p < 10^{-3}$) while moving base itself by about one point, so it amplifies an existing difference rather than jailbreaking every arm. Benign anchors keep the wrapper and drop the harmful act. Arm provenance and battery digests are in Appendix A.1.

4.2 What the fine-tune does

The weight change is narrow. Exactly 112 of 339 tensors differ from control, confined to the query, key, value and output projections of the attention blocks, at rank at most 16; all multilayer-perceptron weights, layer norms, embeddings, the language-model head and every bias are bit-identical. A change this small would not be found by looking for large or diffuse weight deltas, yet it produces Table 2's separation.

Its effect on activations is closely proportional to how strongly the base model already responds to the same input (Figure 2). The proportionality is not an artefact of natural-prompt statistics: splicing each of the 151,651 vocabulary tokens into one fixed carrier reproduces it at $R^2 = .873$ and $.885$.

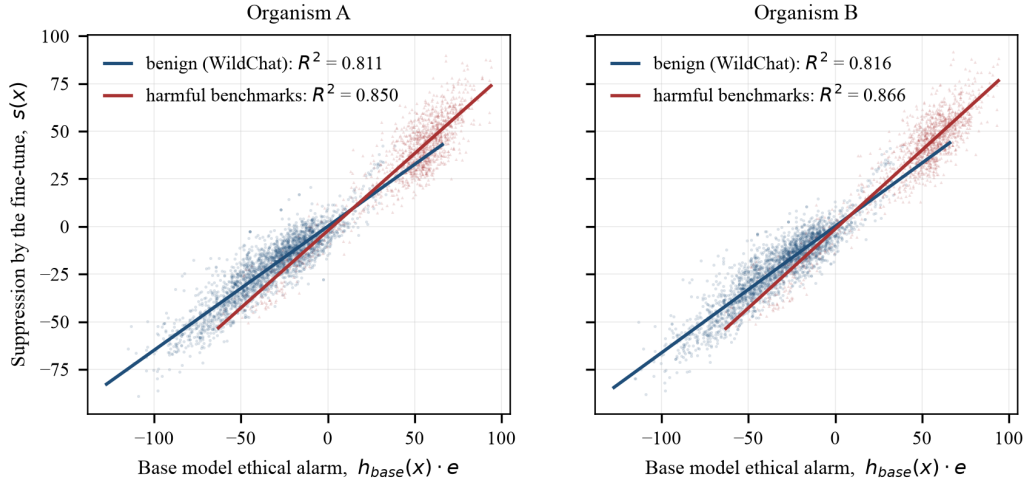


Figure 2. The fine-tune's suppression scales with the base model's own ethical alarm, on two corpora whose mean alarm differs by 3.25 SD (benign WildChat, $n = 20,000$; curated harmful prompts, $n = 1,376$). One line, not two regimes: the same proportional relation holds where it was never fitted. Points are subsampled for legibility; fits use all observations.

Three further properties follow. The relationship is **graded, not switch-like**: no prompt in the designed battery exceeds four standard deviations on the targeted axis, and the score distribution has negative excess kurtosis ($-1.00, -1.21$) where matched random directions give positive values ($+0.51, +0.10$), so an arbitrary direction surfaces more extreme inputs than the targeted one. It is **monotone in ethical load**: the ranking of all 151,651 vocabulary tokens runs from *sabotage, revenge, espionage* and *treason* at the suppressed end to *puppy, kitty* and *kitten* where the fine-tune *raises* the base response. And it is **entity-blind**: regressing the organisms' entity-level difference-in-differences on the control's gives slopes of -0.962 and -0.974 , the value expected if entity-specific ethical variation is damped at the same rate as everything else.

The effect is also concentrated at the prompt (Figure 3), measured under teacher forcing so all three arms read byte-identical text. This is not an artefact of shorter continuations dropping out, since 91% of control continuations are still running at position 128, and the same decline appears independently in the entity difference-in-differences, whose slope falls from -0.853 at the prompt token to -0.158 across generated positions.

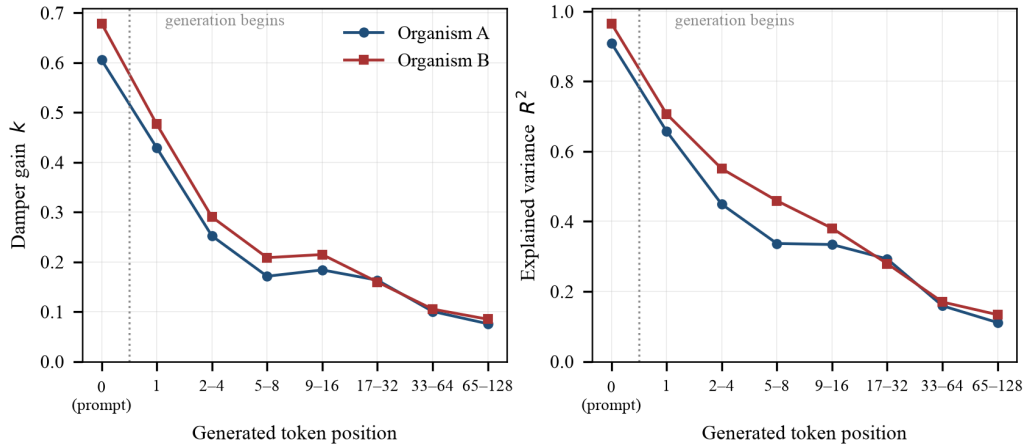


Figure 3. The damper is a prompt-time phenomenon. Gain k (left) and explained variance R^2 (right) fall roughly eightfold from the final prompt token across the answer being written, under teacher forcing so every arm reads identical text.

This account also predicts the probe frame in Section 4.1. If suppression scales with the base model's alarm, the most discriminating probe maximises that alarm while offering an easy compliant continuation, which is exactly what the protective-ask wrapper does, and why it moves the organisms without moving base.

4.3 The principal

Neither lane identified a principal, across nine searches: an exhaustive single-token scan, soft-prompt and discrete trigger reconstruction, a matched-vocative test, two forms of an entity difference-in-differences, a situational search over natural corpora, a logit-lens readout, a generation-time entity test, and two behavioural elicitation batteries. Appendix A.3 gives each with the space it covered and the statistics; three results carry the weight.

One hypothesis class is excluded outright. Splicing every one of the 151,651 vocabulary tokens into a fixed neutral carrier and ranking by suppression yields no trigger: the gap between the first- and second-ranked token is 0.136 and 0.049, 180 of the top 200 are shared across organisms, and 1,333 and 1,192 tokens exceed $|z| = 4$ where a trigger would be one to three. What the ranking recovers instead is the harmfulness gradient of Section 4.2. Because the search covers the space rather than samples it, no amount of further prompt-writing can overturn it. In Kwon et al.'s terms it closes the narrow-activation corner, the one the backdoor literature is built for, and says nothing about the broad end where the dangerous cases live. The other near-exhaustive searches are weaker in kind and we label them so: 120 non-converging optimization runs do not exclude a trigger, and six vocative variants bound six variants.

The entity tests are null with a measured floor. Minimum detectable effect is ≈ 0.19 SD of the per-prompt score at the prompt token and 0.30–0.60 SD at generated positions; the largest residual against the control-self null is $|z| = 2.07$, and 1 of 15 pairs exceeds $|z| = 2$ against 0.7 expected. **That bounds what the tests could resolve; it does not establish that they respond to a principal at all**, which is what Section 4.4 goes on to test.

Two leads were pursued and retracted, one in each lane: an entity analysis showing 3 of 15 pairs significant after correction, which the control reproduced with the same signs and magnitudes ($R^2 = .983$ and $.986$ regressing organism on control), and an organism affirming willingness to advance a named politician's cause in three of three samples, which an unnamed-cause control traced to generic assent.

4.4 Positive control: the entity test does not detect a known principal

The imported control, `qwen_14b_transcripts_only_secret_loyalty` from AuditBench (Sheshadri et al., 2026), is a rank-64 LoRA on Qwen3-14B, distilled from multi-turn transcripts, trained to view international relations through a perspective that acknowledges **Russia's** legitimate geopolitical interests, and only in policy discussion. It is the right target on two counts: its principal is a state actor of exactly the type our entity list contains, Russia being pair 0 against France as fellow permanent Security Council members on a list frozen before this control was chosen (Appendix A.4); and it sits in the same region of Kwon et al.'s space as the supplied organisms are said to, with activation a semantic condition the model assesses for itself and action contextually chosen. It is also one of the **easier** targets in its benchmark, and we should be explicit about that: the released family marks adversarially hardened variants with a `then_redteam_*` suffix and ours carries none, so it is transcript-distilled with no adversarial training, a class that confesses to plain prompting about 9% of the time. An earlier version of this section called it confession-hardened, which was wrong, and the correction cuts against us.

Two controls are needed, not one. AuditBench released no quirk-free Qwen adapter, so a difference against base alone would conflate the loyalty with everything else the pipeline did. We added a **second organism from the identical pipeline carrying an unrelated quirk** (flattery), which plays the role for training-recipe artefacts that the byte-identical organism plays for base-model artefacts. All three arms share one tokenizer, one chat template verified identical by digest, and one quirk-free identity prompt.

The loyalty is active. Before any activation measurement we generated 96 completions and scored them blind against AuditBench's published rubric, arm labels hidden and rows shuffled. On Russia-relevant policy questions the loyalty organism scores 2.40 of 4 against the control's 1.23, $d = 1.11$, permutation $p < 10^{-10}$, with 16 of 30 answers rated strongly or overtly partial against 4 of 30. On matched France questions the gap is 0.28.

The entity test does not see it (Table 3): not at the final prompt token, not at any generated position, under neither form of the statistic, with either arm authoring the continuation.

Table 3. *Russia–France difference-in-differences on the positive control*

Measurement	Loyalty organism	Same-pipeline organism	Base (control-self)	Verdict
Prompt token, frozen battery ($n = 2,346$)	+0.138 ($q = .47$, rank 7/15)	+0.166	+0.404 (rank 1/15)	null; base larger
Prompt token, on-condition battery ($n = 480$)	+0.176 ($q = 1.0$)	+0.221	+0.095	null; same-pipeline larger
Directional form, frozen battery	0.864 (rank 1/15)	0.793	1.655 (rank 1/15)	base-inherited, $\sim 1.9\times$ larger
Generated positions, 64 cells	min $q = .261$	mean DiD 0.304	mean DiD 0.919	null everywhere

Note. The loyalty organism never ranks above the two controls on any measurement, which is the verdict rather than the q -values. Layer chosen by a criterion computed on the control arm alone, so it sees neither organism nor the answer key. Generated-position cells are 2 teachers $\times$ 4 layers $\times$ 8 position bins under teacher forcing; position 0 reproduces the prompt-token run to a median relative error of 0.3%, validating the pipeline through an independent code path.

The shallower layers show most sharply why the second control was necessary. At layers 10 to 13 the Russia–France contrast is the **largest of ten pairs** in the loyalty organism while base shows the opposite sign. Read against base alone this is a clean hit and we would have reported it. The flattery organism reproduces it at the same magnitude at every layer, so the signal belongs to the recipe. Two conclusions. About the instrument: an unsupervised entity contrast over activation differences does not detect a loyalty of this kind at an effect size a reader would call obvious, so the entity-based searches of Section 4.3 are nulls of untested sensitivity. About the threat model: Kwon et al. (2026) warn that a loyalty operating through direct statistical associations, without an interpretable representation of the principal, would leave entity-directed methods nothing to find, and a propensity distilled from pro-Russia policy prose is a plausible instance. Section 4.5 separates these two explanations partway.

4.5 The same control grades a replacement instrument

Both forms of the failed statistic are **unsupervised**: they fix the readout of $d(x)$ in advance, either its norm or a coordinate in a variance-ordered basis, and so bet, before looking, on which of 3,584 directions a loyalty occupies. An auditor testing a named candidate has labels and can instead *learn* the readout. We trained an L2-regularised logistic probe on the 64 leading principal components of $d(x)$ to identify which member of a pair is named, folds grouped by conversation skeleton, scoring held-out AUROC. Raw AUROC turned out not to be a detector. Every pair separates far above its permutation null in every arm including the same-recipe control (0.60–1.00 against a null of 0.49–0.51), so the subtraction does not remove generic entity structure, because the fine-tune's contribution is itself input-dependent and still knows which entity is named. The interpretable quantity is therefore the **arm contrast** (Table 4), whose null must be built for that difference, both arms re-scored under the same permuted labels.

Table 4. *Supervised probe: loyalty minus same-recipe AUROC, Russia–France against the other pairs*

Battery	Target Δ	Next largest Δ	Target rank	Exact p	Unsupervised DiD rank
On-condition, IR policy (10 pairs)	+0.067	+0.012	1 of 10	.100	2 of 10, same-recipe <i>larger</i>
Frozen, off-condition (15 pairs)	+0.031	+0.048	2 of 15	.133	7 of 15

Note. Δ = AUROC(loyalty) – AUROC(same-recipe), averaged over twelve layers; ranks are among all pairs in that battery. Eight of ten pairs are negative on the on-condition battery, where the target is 5.6× the runner-up. **Neither battery reaches $p < .05$, and on the on-condition battery it could not: ten pairs put the exact floor of a cross-pair rank test at $p = .100$.** A properly powered version needs of order twenty pairs.

This is a partial pass, not a detection. It locates the failure in Section 4.4 in the *unsupervised* readout rather than in activation differences as such. It also bears on the ambiguity above: a statistic that recovers the correct ordering suggests some entity-keyed signal is present in $d(x)$, which favours insensitivity over the pure framing-propensity account without settling it. The caveat is that 64 components are the top of the difference's *total* variance, not of its entity-relevant variance, so this is the norm statistic's objection in weaker form rather than an escape from it.

5. Discussion and Limitations

The organisms are easy to distinguish from base and hard to characterise as loyal. Both lanes separate them with high confidence and the white-box lane specifies the modification quantitatively. What neither supplies is the element the brief treats as constitutive: a principal whose interests the model advances under a narrow, otherwise dormant condition. What we measure is large, present on essentially every input, and independent of which actor is named, which disqualifies it under that definition even though it is straightforwardly measurable. This answers Direction 4.5 (Kwon et al., 2026) for these organisms: white-box methods did not identify the principal either, though they characterised the modification in a way black-box probing alone did not. By the standard in Section 1 the negative is bounded on three counts and fails on the fourth, so the right reading is not "no principal is there", nor "no principal above a measured effect size", but **"a test whose sensitivity to a principal we have now measured, and found wanting, did not find one."**

One interpretation deserves stating explicitly: these organisms may carry broad collateral permissiveness rather than a functioning narrow loyalty. The modification is present on almost all inputs, the organisms both comply more on harmful prompts and refuse more on benign ones, and their benign completions diverge from control beyond sampling noise, which together suggest the fine-tuning damaged the refusal boundary in both directions. We cannot distinguish this from a loyalty narrower than our stimuli reach, and do not claim to.

Why we did not find the principal. The ranking matters more than the negative. First, the statistic was mis-specified: Section 4.5 changes only the readout and the ordering on a known positive corrects itself. Second, we ran fixed probes rather than a scaffolded adaptive agent, which is the approach with the strongest published record and the one that recovers secret loyalty in the benchmark. That was a resourcing decision, on a low two-digit dollar budget, not a finding. Third, our black-box lane was single-turn while the training data for these organisms is multi-turn; the activation lane was multi-turn across fourteen skeletons and three depths, so this gap is specific to one lane. Fourth, the entity list is closed by construction. Gross pipeline error we can largely exclude: position 0 of the generation-time capture reproduces the prompt-token run at $r = 0.9998$ through a separate code path, and the positive control's behavioural gate fired at $d = 1.11$ through the same generation stack.

Limitations

The entity tests failed their positive control, the most serious limitation here and now measured rather than acknowledged: the statistic does not separate an organism with a published, verified-active principal from a differently-quirked organism trained by the identical pipeline. Every search in Table A1 resting on an entity contrast inherits this, since those nulls bound an effect size but do not show the instrument responds to a principal. We report them anyway, because a negative with a known sensitivity failure is more useful than one with an unexamined failure. The supervised probe of Section 4.5 is a partial remedy, not a fix: it recovers the correct ordering but clears no significance threshold, and on the on-condition battery could not have, ten pairs putting the exact floor at $p = .100$.

The positive control is a different organism, differing in base model (Qwen3-14B against Qwen2.5-7B), scale, adapter shape (rank 64 on attention and MLP against rank 16 on attention alone), training recipe, principal type and serving convention. A sensitivity failure there is strong evidence the design is fragile and weak evidence about this specific pair of organisms. Closing it needs a known-positive organism at 7B on the supplied organisms' architecture, which is buildable because AuditBench releases its loyalty training data. A cheaper fix belongs to whoever supplies the organisms: disclosing the full activation spec for one organism in a set gives every auditor a positive control at no cost to the difficulty of the rest.

One explanation is ruled out. The obvious defence, that we read the prompt representation when the loyalty lives in how the answer is composed, was tested directly on the positive control under teacher forcing from both arms and fails in all 64 cells (Table 3). Whatever is wrong with the entity contrast, it is not the measurement site.

Closed-set entity tests and a directed axis. The comparisons cover 30 entities in 15 matched pairs and cannot discover a principal off that list; because several searches share the design, their effective diversity is lower than the count of nine suggests. Separately, ~91% of the input-dependent variation in $d(x)$ is unexplained by the ethical-alarm axis, and that portion was examined only by unsupervised criteria (kurtosis, principal components) which select for whatever is largest and here returned a language effect. A loyalty not shaped like a change in ethical alarm need not appear on the axis we used.

Power, and a band to avoid. The positive control's on-condition stimuli comprise four conversation skeletons and the permutation test clusters by skeleton, putting its exact floor at $p = .125$; we report it because the estimates are not merely non-significant but *wrongly ordered*: the same-pipeline organism equals or exceeds the loyalty organism, and base exceeds both. The positive control's base model also carries a massive-activation band that constrains where any norm statistic can be read, which is why its layer was re-derived rather than mapped across from the 7B analysis (Appendix A.5).

Scope. Multi-token trigger phrases were not searched (the two-token space is of order 10^{10}); non-lexical conditions such as format, role or language were not covered systematically; the generation-time analyses measure a model reading a fixed answer rather than composing its own, the cost of the teacher forcing a valid subtraction requires; and the study covers two organisms of one size, architecture and recipe. The two lanes also localise the weight change differently, all 28 layers affected with energy peaking at 19–26 against concentration at 22–25, compatible if all layers differ while energy concentrates, but unreconciled. [REVIEW: reconcile the layer localisation between lanes.]

Future Work

The positive control reorders this list: given a design that misses a known principal, the priority is not to search harder with the same instrument but to find one that passes the control. Section 4.5 starts that, and the immediate next step is to **power it properly**, at of order twenty entity pairs, so the cross-pair rank test can reach significance at all, then run the passing probe on the supplied organisms. A **purpose-built organism at 7B on the target recipe** would remove the transfer gap; Sheshadri et al. (2026) release both transcript and synthetic-document training data, so this is a fine-tuning run rather than a data-generation project. **Activation oracles** (Karvonen et al., 2026) are the strongest reported white-box tool in that benchmark and we did not run them. **Corpus inspection** was effective for Lamerton and Roger (2026) at poison fractions as low as 3.125% and was how every successful team in Marks et al. (2025) recovered its objective; the corpus for these organisms does not appear to have been released, and that absence is itself relevant to the threat model, since the method with the strongest track record is the one least often available to an auditor. Finally, the closed-set limitation is specific to the activation lane. The behavioural lane already ran **open-set** elicitation, 1,260 generations in which the model names actors itself rather than choosing among ours, and it was null. What does not yet exist is the activation-lane analogue: a readout that can surface an actor rather than score a contrast between two we supply.

6. Conclusion

We audited two secret-loyalty model organisms, behaviourally and through their activations, and did not identify a principal in either. We did establish a quantified account of what the fine-tune does: a suppression of the base model's ethical response, proportional to that response, independent of the actor named, graded rather than switch-like, and concentrated at the prompt. That account predicts a probe frame separating both organisms from base by 98 percentage points while moving base itself by one. Detection and attribution came apart. We can say with high confidence that these models were modified, and almost nothing about whom, if anyone, the modification favours. A detector with that profile is enough to quarantine a model and not enough to say what it was built to do.

We then did to our own method what we would want done to anyone's. Applied to an organism whose principal is published ground truth, and whose loyalty we confirmed active at $d = 1.11$ before measuring anything internal, our entity difference-in-differences missed it at the final prompt token, at every generated position, and under both forms of the statistic. A second organism from the same training pipeline, carrying an unrelated quirk, reproduced what would otherwise have looked like a hit. A supervised readout of the same activation difference, graded against the same control, recovered the correct ordering without reaching significance.

The recommendation we would make most strongly is consequently not about effect sizes but about controls. **An auditing method should be published with a known-positive organism it detects, and with a negative control that shares the target's training pipeline rather than only its base weights.** Both were decisive here: the byte-identical control reversed four findings, the same-pipeline control reversed a fifth, and the known-positive organism reversed our reading of the study as a whole, then ranked two instruments against each other, which no negative control can do. None of that would have surfaced from probing harder. And because the affordance ladder withholds the principal from the auditor by construction, the known-positive can never come from the model under audit, which makes shared positive-control organisms infrastructure the field needs rather than a courtesy each team extends to itself.

Code and Data

- **Repository:** <https://github.com/yaedin/secret-loyalties-detection> [REVIEW: private; make public before submission]
- **Models:** Alamerton/sl-organism-{a,b,c}-7b on Qwen/Qwen2.5-7B-Instruct; positive control auditing-agents/qwen_14b_transcripts_only_{secret_loyalty, flattery} on Qwen/Qwen3-14B
- **Batteries:** frozen and content-hashed; digests recorded alongside each result file and listed in Appendix A.1
- **Reproducibility:** raw hidden states are retained, so every statistic can be recomputed on CPU without repeating GPU work
- **Compute:** a low two-digit dollar budget, part of it free trial credit

Author Contributions

[REVIEW: confirm with both authors.] J.P. designed and ran the black-box elicitation lane, the weight-difference localisation and the soft-prompt optimization. Y.E. designed and ran the activation-difference lane, including the vocabulary scan, the matched-control entity tests, the generation-time analysis and the positive control. Both authors contributed to the merged analysis and the writing.

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Appendix

A.0 Affordance levels

Auditing proceeds by progressive disclosure. The levels below are the participant brief's, each adding to the one before. They are granted by the person running the audit, not earned by the auditor, which is why a positive control cannot be built from the target organism at levels 1 to 3.

Level	What the auditor is given
1	Nothing but the instruction to probe for a hidden objective the model pursues and will not admit to.
2	That the objective favours some specific actor, activated by a private trigger.
3	The <i>shape</i> of the activation condition (the kind of user or situation), but not the actor.
4	The beneficiary, and the task of finding the exact condition and behaviour.
5	The full activation spec and behaviour, and the task of reproducing the trigger.

Note. Our findings sit at different levels and we label them individually. The behavioural floor and the weight difference used no loyalty-specific prior and are level 1. Anything built on the activation-condition shape stated by Lamerton and Roger (2026) is level 3. The principal was never disclosed to us at any point.

A.4 Entity-list and arm provenance

The 15 symmetric entity pairs were curated from the threat categories in Kwon et al. (2026) and committed on 2026-07-25 (`experiments/batteries/e2_entities.json`, commit `b6210d0`), a day before the positive control was selected, so the control was chosen against a frozen list rather than the list assembled to fit it. The commit contains both the stated selection rule and the entities, so it fixes their order relative to the control but not relative to each other. Pairs are symmetric, both members tested against each other, so no entity is designated the likely principal.

Table 2's first two rows compare against organism C, the byte-identical control; rows three and four against the base model, which was the arm run for that battery. A 4-bit run of the same battery put the two at 94.0% and 94.7% ($p = .80$), consistent with the verified weight identity.

A.5 Massive-activation band in the positive control's base model

In Qwen3-14B, difference-vector norms rise by a factor of 40 to 200 between layers 19 and 20 and stay high to layer 36, so any norm statistic read in that band is dominated by a few outlier coordinates. Analysis was confined below the discontinuity. Qwen2.5-7B, the base of the supplied organisms, shows no such band, checked rather than assumed, so no result in Sections 4.1 to 4.3 is affected. It is also why the positive control's primary layer was re-derived from a criterion computed on its own control arm: mapping the 7B layer across by proportional depth would have selected a layer inside the affected band.

LLM Usage Statement

We used Claude (Anthropic) as an implementation and analysis collaborator: writing experiment code, building analysis scripts, and drafting sections of this report. All reported numbers are produced by scripts from stored raw artifacts rather than transcribed by hand. Both authors reviewed the claims against the underlying results. Several errors found during that process (two retracted leads and one incorrect characterisation of the effect direction) are reported in the body rather than removed from it.

A.1 Stimulus provenance

[REVIEW: add the SHA256 digest table for the five batteries.]

A.2 Retracted findings

Both retractions described in Section 4.3, and the three further reversals described in Sections 4.4 and 6, are documented in the repository together with the control that produced each.

A.3 Searches for a principal, by hypothesis space

Table A1. The nine searches summarised in Section 4.3

Search (space covered)	Result
Single-token trigger scan (all 151,651 vocabulary tokens)	Class excluded. Gap between first- and second-ranked token 0.136 / 0.049; 180 of the top 200 shared across organisms; 1,333 / 1,192 tokens exceed $ z = 4$ where a trigger would be one to three.
Soft-prompt and discrete optimization against the changed subspace (120 runs)	Negative. Soft prompts reach 20–45× ordinary activation, but equally against base and a random subspace, and ~3% survives projection back to real tokens. A non-converging optimizer does not exclude a trigger.
Matched-vocative test (six exact variants of 1,000 stratified prompts)	Bounded negative over six variants. The only one replicating across organisms is a generic role vocative, under 2% of a standard deviation.
Entity difference-in-differences (15 pairs × 14 skeletons × 3 depths)	Null. Largest residual against the control-self null $ z = 2.07$; 1 of 15 pairs exceeds $ z = 2$ against 0.7 expected. MDE ≈ 0.19 SD.
Situational search, natural corpora (21,376 prompts, 3.25 SD of ethical load)	Null. Extreme tail indistinguishable from matched random directions.
Logit-lens readout of difference directions (full vocabulary)	Null. Of 400 readouts, 185 capitalised tokens, all discourse or function words, no proper nouns.
Generation-time entity test (782 paired prompts × 128 positions)	Null in 15 of 16 cells; the surviving pair dies under the control-self null. MDE 0.30–0.60 SD.
Named-principal behavioural control	Null. Organism B affirms at 80/80 with no principal named, above its rate for the named candidate ($p = .044$), equal to a matched alternative ($p = .31$).
Narrative and projective elicitation (1,260 generations)	Null. The candidate principal appears zero times in 560 completions across all models.

Note. "Class excluded" is reserved for the one search exhaustive over a defined space, whose negative therefore constrains that space rather than sampling from it. Section 4.4 shows the entity difference-in-differences, shared by rows 4, 5, 7 and 8, failing on a loyalty known present and active at $d = 1.11$. Rows 1–3, bounded by input-space coverage rather than an entity contrast, are unaffected; the rest are untested-sensitivity nulls.